

CUI

CONTRACT NUMBER (Given by CONS when awarded)

Dated: x date xx As Negotiated date of contract award

STATEMENT OF WORK

SURVEY/DESIGN ANALYSES TO REPAIR, UPGRADE, OR REPLACE CAS SYSTEM AT DYESS AFB, TX

AS NEGOTIATED

DATE

*this will be added by CONS once awarded

DRAFT

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SURVEY/DESIGN ANALYSES TO REPAIR, UPGRADE, OR REPLACE CAS SYSTEM

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TABLE OF CONTENTS

SECTION	PAGE
1.0 General	3
1.1 Scope	4
1.2 Notice to Proceed	5
1.3 Government Furnished Equipment/Materials	6
1.4 GSC and Base User	6
1.5 Task and Deliverables	6
2.0 Project Management	8
2.1 Government Points of Contact	8
2.2 Contractor's Project Manager	8
2.3 Payment Monitoring Plan	9
2.4 Schedule	9
Appendix-A	10

STATEMENT OF WORK (SOW)

SURVEY/DESIGN ANALYSES TO REPAIR, UPGRADE, OR REPLACE CAS SYSTEM

1. GENERAL

The Centralized Aircraft Support System (CAS) at Dyess AFB, Texas is approximately 30 years old. The components and infrastructure of this system have not been inspected or utilized since 2013 and their current condition is unknown, thereby making the system 100% non-operational due to safety and environmental concerns. As a result of the unknown condition of the CASS components and associated infrastructure, Aerospace Ground Equipment (AGE) has been the primary means of supporting flight line aircraft maintenance operations. A recommendation of specific requirements, to include materials, equipment, and labor, for modernization or replacement of the CAS system will allow the government to make the most cost effective decision relative to upgrading or replacing this system at Dyess AFB. This study will provide detailed requirements and justification for project scope. The study will include a survey of content that includes itemized list of work required, equipment/component replacement cost estimate, and manufacture/vendor recommendations, with emphasis on a long term sustainment program and life cycle management costs. Replacement of system components and/or infrastructure must meet or exceed existing performance specification requirements when fully operational.

The CAS system has five remote substations and 21 individual CAS system pit parking spots; each individual parking spot contains five CAS system pit access points that house five conditioned air system hoses, one electrical system power line, two coolant system recirculation lines, two bleed air supply lines, and one fuel servicing line. The five remote substations provide electricity and conditioned air to each CASS pit. Main power is fed from a 480V/2500A 60Hz platform via Substation 2. Recent construction has demolished and covered 9 CAS system parking pits (Positions A 1-5 and B 2-5). Only 3 Loops remain each feeding to 4 Pits.

Each of the 12 designated CASS pit locations were designed to provide conditioned air at a minimum of 45 degrees Fahrenheit delivered up to 700 lb/min at 42.9 btu/lb at 53 in H₂O to keep aircraft internal temperatures at 80 degrees or lower during full operation. Each pit also provides three phase 230/400VAC, 400 Hz, 105/115 KVA external electrical power used to facilitate scheduled and unscheduled aircraft maintenance. The coolant system delivers pressurized coolant oil to the aircraft liquid cooling system with a cooling capacity maximum of 200 lb/min providing 37 btu/lb at 158 psig (110 psi minimum) and 80 GPM max (nominal 55 GPM). The Dielectric Coolanol Coolant (MIL-C-87252C) is used for advanced operational system checks. The coolant oil system is not a mandatory section of the CAS system; however, the Government is seeking input from commercial industry on the most cost effective solution to

repair this system; all repair inputs are welcomed. Lastly, each spot is fitted with a bleed air source to supplement or replace the Dash 95 air carts required for bleed air trouble shooting and additional cooling of the environmental system. This system is pressured using a turbine engine with an output of 147 lb/min at 48 psia, the bleed air system is not a mandatory section of the construction/repair of the CAS system; however, the system will need to be removed.

The fuel servicing line is maintained by the Logistics Readiness Squadron (LRS), and will not be addressed in this report.

All recommendations shall include life cycle costs, and an estimate return on investment. An asbestos report shall be included if applicable.

1.1 SCOPE

1.1.1 GENERAL REQUIREMENTS: The Contractor shall furnish all labor, materials, equipment, and transportation required to provide a survey/study of the CAS system of what it will take to restore, replace, or upgrade the CAS system at Dyess AFB, reducing the overall manpower needed to conduct maintenance on the B-1 fleet. The Government intends to utilize the resulting deliverables from this contract to develop future repair/revitalization courses of action. The 400Hz power and distribution system needs to be upgraded. The 28VDC power that is used for the entire CAS system control circuit requires upgrading. The Polyvinyl Aircraft Coolant (POA) oil system needs to be environmentally isolated and deactivated. The chillers and blowers need to be surveyed for expected life cycle replacement. All pit doors and locking mechanisms need to be surveyed for operability. All pit drainage systems need to be surveyed for effectiveness. Work shall be accomplished in accordance with the references listed in Appendix A and applicable publications, except as modified elsewhere in this SOW.

1.1.2 400HZ AIRCRAFT POWER: The power and distribution system requires upgrading and/or replacement. This system is 30 years old and replacement parts are unavailable due to the technology of the original design being obsolete. The entire system is inoperable and is currently replaced with portable AGE to support flight line aircraft maintenance operations. The Contractor shall provide a detailed analysis of existing components and infrastructure, as well as any recommendations for repairs, upgrades, or replacement with a new 400Hz power and distribution system which meets or exceeds all codes and aircraft requirements. The contract deliverables for this study is a detailed listing of existing and replacement equipment (if recommended), cost estimate (materials & labor) for repairs, upgrades, and replacement, and a listing of manufactures/vendors that can provide service and parts after contract award, with an emphasis on a long term sustainment.

1.1.3 28 VOLT DC (VDC) POWER: The 28VDC power system is used for the entire CAS system control circuit (excluding liquid fuel system) and is supplied from the 400Hz power distribution center. Loss of the 400Hz Converter results in loss of the entire CAS system.

The Contractor shall provide operational diagrams and a recommendation for changes to the current system configuration which will allow maximum flexibility to accomplish separation of the 28VDC control circuit from the 400Hz power system, complete with a general statement of work, cost estimate, and list of manufactures/vendors which can provide service after contract award, with an emphasis on long term sustainment.

- 1.1.4 LIGHTNING AND EXPLOSION PROTECTION:** Analyze existing lightning protection/surge suppression and explosion protection capabilities for current and future system requirements. Provide recommendations for surge protection/suppression and explosion protection, as required in an aircraft industrial environment.
- 1.1.5 AIR COOLING:** The Air Conditioning system needs to be looked at in several areas. Provide name and location of manufacturer and cost of all upgraded equipment that meets or exceeds present performance on equipment specifications, with an emphasis on a long term sustainment program. Provide source and costs for upgrades on current electrical relays and subcomponents that are obsolete. Provide chiller and blower replacement specifications, design, and costs, promoting ease of maintenance and operation.
- 1.1.6 PAO COOLING (LIQUID COOLING):** The Polyvinyl Aircraft Coolant (PAO) cooling oil system will be disabled and will require evaluation to determine if some or all existing external components need to be removed and/or replaced. The government is requesting the system be drained and cleaned out so that the imbedded piping may then be used to run future conduits for electrical system upgrades. Cost of draining and cleaning will be added into the recommendation. This will assist in transitioning from pneumatic to electrical control power of the air delivery system. If determined not to be cost effective, the Contractor will provide various option, the estimated costs, and the life cycle cost for system upgrades that meets the same intent.
- 1.1.7 LINE EQUIPMENT CAVITY UPGRADES:** The storage pits imbedded in the parking apron are in need of repair, upgrade, or replacement. The lids and locking mechanisms on all of the pits are broken or of poor design as they do not lock, which poses a potential driving/tripping hazard while working around CAS pit areas. The drainage system of the individual pits are inadequate and require cleaning or upgrade to allow proper drainage of water/fuel away from electrical systems. The entry/exit ladders of the pits are damaged and need to be repaired. The illumination systems of the pit cavities are disabled and request return to service. The contractor shall provide recommendation for upgrading and/or replacing the storage pits to include the estimated costs and the life cycle costs to maintain. If there is other cost effectiveness ways that meet the intent of the line equipment cavity upgrades, the contractor shall make those recommendations and the associate life cycle costs.

1.1.8 SYSTEM MONITORING: Provide recommendations for the installation of a standalone monitoring system that would enable quick review of system parameters and operational capabilities.

1.2 NOTICE TO PROCEED: Upon receipt of this delivery order the Contractor shall commence work on Task 1, Project Work Plan.

1.3 GOVERNMENT FURNISHED EQUIPMENT/MATERIALS

1.3.1 MATERIALS AND WORKSPACE: In advance, at least 20 days, of the one-week Data Gathering and Analysis Charrette, the government will provide as-built drawings of equipment and pit locations, and a base map of Dyess AFB. The government shall provide a suitable work space (preferably a conference room) for the Charrette Team to use during the Charrette week.

1.3.2 ESCORTED ACCESS: The government will provide access to the equipment on or off the flight line. Contractor shall work with 7 CONS and the Escort POC to schedule escorts. A camera pass may be provided by the government in advance of the Charrette week if requested by the Contractor. The process for requesting a camera pass will be provided to the Contractor at the post award conference and/or emailed to the contractor award of this contract.

1.4 GSC AND BASE USER

The Contractor shall meet with designated base Contracting (CONS), CES, and MXG representatives at Dyess AFB, which will be the end user of the final product, at the kick off meeting for the Data Gathering and Analysis Charrette. During this meeting the Contractor will present their time line for the survey and specific needs. The AFGSC/A7CI Technical Project Manager (TPM) will also attend the meeting, if able.

1.5 TASK AND DELIVERABLES

Task-1: Project Work Plan

The Contractor shall produce and submit a project work plan for approval. The plan shall be completed in Microsoft Project, and contain at a minimum, the strategy and milestones for completing the project task (Deliverable 1). Deliver three paper copies and an electronic copy of the Project Work Plan (two copies to the Dyess AFB Project Manager (PM) and one copy to the AFGSC A7CI for approval within 14 days of Notice to Proceed (NTP).

Task-2: Coordination of Meetings to Start Field Surveys

Following NTP, the Contractor shall schedule a coordination meeting at the beginning of the one-week Data Gathering and Analysis Charrette at Dyess AFB with the Base Project Manager (PM), Contracting Officer/Contracting Specialist, and Engineers from the CES to review scope tasks and procedures. The one-week Data Gathering and Analysis Charrette

shall be scheduled far enough in advance to provide the Contractor more than 15 days prior to the Charrette to purchase airfare and obtain lodging. The dates of the Charrette may not change within 15 days of the event without a modification to the contract to compensate for the increased airfare cost. The Contractor shall submit meeting minutes to 7 CONS on Dyess AFB. The Contracting Officer will distribute the copies appropriately to the Dyess AFB Project Manager, 7 CES and AFGSC A7CI Project Managers (Deliverable 2). The Contractor shall conduct necessary field surveys to collect the data and verify data furnished by base PM or other organizations.

Task-3: Field Survey Exit Briefing

The Contractor shall meet with the Base PM, CES, CONS and other representatives to present general findings of the field surveys and to discuss the impressions resulting from the initial phases of the engineering evaluations and system analyses. The Contractor shall discuss anticipated problems and strive for clarified direction if portions of the study tasks appear questionable or unsound as a result of the fact-finding and initial evaluation efforts. The Contractor shall submit meeting minutes to 7 CONS on Dyess AFB. The Contracting Officer will distribute the copies appropriately to the Dyess AFB Project Manager, 7 CES and AFGSC A7CI Project Managers (Deliverable 3).

Task-4: Draft Report

The Contractor shall prepare a draft report including the results of all analyses of the CAS system inclusive of the power, controls, and mechanical HVAC equipment. An executive summary of the study recommendations shall be provided as part of the report for easy reference. The recommendations shall provide an executive summary of the costs broken down by system component (air, 400Hz power, and oil coolant systems) and the estimated life cycle costs and estimated return on investment.

As part of the "draft report", the Contractor shall submit cost estimates, performance parameters, interface requirements, list of equipment, list of sources, photographs or digital images showing where possible all field conditions influencing conclusions in the submittal. A narrative report shall be submitted describing the general conditions and all specific conditions for each photograph. All photographs shall be submitted with design analysis and will clearly show field conditions and verify quantity and quality of work required. Provide all photos on CD-ROM using JPG format. Three CD-ROM of the photographs will be submitted to 7 CONS on Dyess AFB. The Contracting Officer will distribute the copies appropriately to the Dyess AFB Project Manager, 7 CES and AFGSC A7CI Project Managers.

The Contractor shall submit two paper copies and one electronic copy of the analysis and recommendations to the Contracting Officer. The Contracting Officer will distribute the copies appropriately to the Dyess AFB Project Manager, 7 CES and AFGSC A7CI Project Managers (Deliverable 4).

The Contractor shall attend a one-day review meeting at Dyess AFB to review the Draft Report and obtain any comments needing to be incorporated into the work. The one-day review meeting shall be scheduled far enough in advance to provide the Contractor more than 15 days prior to the Charrette to purchase airfare and obtain lodging. The dates of the review meeting may not change within 15 days of the event without a modification to the contract to compensate for increased airfare cost. The Contractor shall incorporate any comments provided by the Dyess AFB Project Manager through the Contracting Officer.

Task-5: Final Report (Deliverable 5)

The Contractor shall prepare the final report incorporating the government's comments on the draft report. The Contractor shall submit two paper copies and one electronic copy of the analysis and recommendations to the Contracting Officer. The Contracting Officer will distribute the copies appropriately to the Dyess AFB Project Manager, 7 CES and AFGSC A7CI Project Managers.

2.0 PROJECT ADMINISTRATION AND MANAGEMENT

The Contractor will work independently with points of contact as identified by the Technical Project Manager (TPM).

2.1 GOVERNMENT POINTS OF CONTACT

Below are the various POCs:

TPM
[TBD]

The Dyess Program Manager
[TBD]

The Contracting Officer
[TBD]

2.2 CONTRACTOR'S PROJECT MANAGER

The Contractor's project manager shall serve as a single point of contact and liaison for all work required under this contract. If the Contractor has not designated the project manager in the proposal or during negotiations, the Contractor shall immediately designate the project manager and an alternate in writing upon award of the contract.

2.3 PAYMENT MONITORING PLAN

The milestones to be used to verify Contractor payment invoices are detailed below:

	Task Percentage	Actual Percent Complete
Task 1 – Work Plan	40%	
Task 2 and 3 – Field Survey & Research	50%	
Task 4 and 5 – Draft Report & Final Submittals	10%	

2.4 SCHEDULE

The Contractor shall complete all required tasks in accordance with the deliverables schedule below. The performance period for this delivery order is 113 calendar days after delivery of all required data on the CAS system, and beginning with the first day of the on-site Charrette.

Project Milestone	Total Task Days	Task Days
Notice to Proceed	0	0
Receive prerequisite date from Dyess AFB	0	0
Coordination with Base on Charrette dates	3	3
Prepare and Submit Work Plan	18	14
Base & GSC Review and approve Work Plan	25	7
Submit revised Work Plan if required	32	7
Kick-Off Meeting and one-week Charrette (including Charrette preparation time)	43	10
Prepare and submit 90% Draft Report	72	30
Base & GSC Review of 90 % Draft Report	87	15
90% Draft Review Meeting at Dyess AFB	93	6
Prepare and submit 100% Final Report	113	20

Appendix A

1.1.2 References: (Electrical Codes and Standards)

- IEEE 3003.1-2019 - IEEE Recommended Practices for Grounding of Industrial and Commercial Power Systems
- IEEE Std 3003.2 – Grounding and Bonding for Industrial and Commercial Power Systems
- IEEE Std 3001.2 – Power Quality for Industrial and Commercial Power Systems
- IEEE Std 3001.1 – Power Systems Design for Industrial and Commercial Facilities
- NFPA 780-2026- Lightning Protection Systems
- ANSI C2-2023- National Electrical Safety Code
- 2026 National Electrical Code

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